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## The Theory of Consumer Choice



# The Theory of Consumer Choice

- The theory of consumer choice addresses the following questions:
  - Do all demand curves slope downward?
  - How do wages affect labor supply?
  - How do interest rates affect household saving?



# THE BUDGET CONSTRAINT: WHAT THE CONSUMER CAN AFFORD

- The *budget constraint* depicts the limit on the consumption “bundles” that a consumer can afford.
  - People consume less than they desire because their spending is constrained, or limited, by their income.



# THE BUDGET CONSTRAINT: WHAT THE CONSUMER CAN AFFORD

- The budget constraint shows the various combinations of goods the consumer can afford given his or her income and the prices of the two goods.
- Suppose there are two goods, Pizza ( $x$ ) and Pepsi ( $y$ ).
- Let Income of consumer is 1000 dollars, price of Pizza is 10 dollars per unit and price of Pepsi is 2 dollars per unit.



- The Budget Constraint in this case is given by
- $10x+2y=1000$

**Figure 1 The Consumer's Budget Constraint: Let Pizza price is 10 dollars per unit, and Pepsi price is 2 dollars per unit**

| Pints<br>of Pepsi | Number<br>of Pizzas | Spending<br>on Pepsi | Spending<br>on Pizza | Total<br>Spending |
|-------------------|---------------------|----------------------|----------------------|-------------------|
| 0                 | 100                 | \$ 0                 | \$1,000              | \$1,000           |
| 50                | 90                  | 100                  | 900                  | 1,000             |
| 100               | 80                  | 200                  | 800                  | 1,000             |
| 150               | 70                  | 300                  | 700                  | 1,000             |
| 200               | 60                  | 400                  | 600                  | 1,000             |
| 250               | 50                  | 500                  | 500                  | 1,000             |
| 300               | 40                  | 600                  | 400                  | 1,000             |
| 350               | 30                  | 700                  | 300                  | 1,000             |
| 400               | 20                  | 800                  | 200                  | 1,000             |
| 450               | 10                  | 900                  | 100                  | 1,000             |
| 500               | 0                   | 1,000                | 0                    | 1,000             |

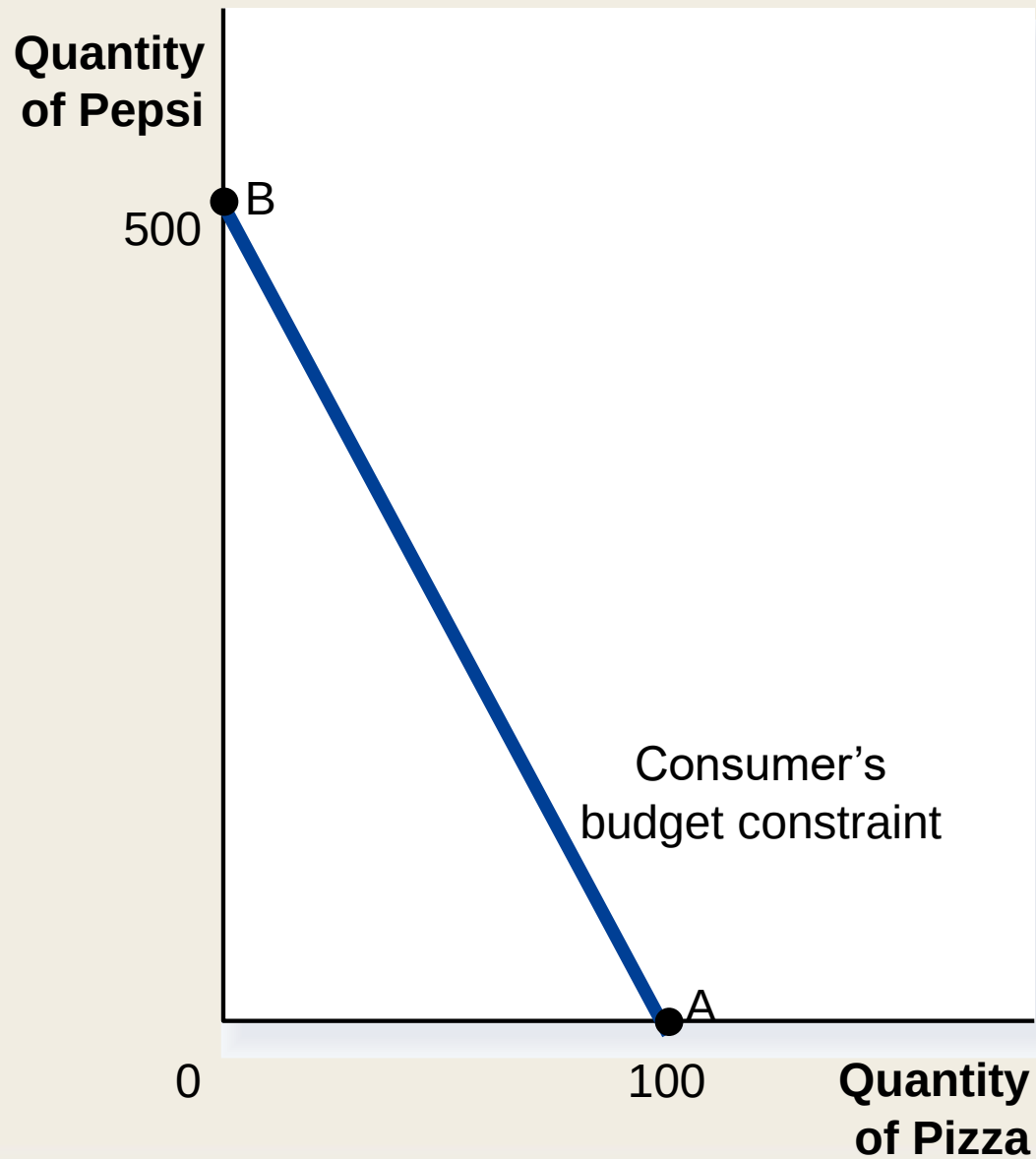


# THE BUDGET CONSTRAINT: WHAT THE CONSUMER CAN AFFORD

- The Consumer's Budget Constraint
  - Any point on the budget constraint line indicates the consumer's combination or trade-off between two goods.
  - For example, if the consumer buys no pizzas, he can afford 500 pints of Pepsi (point B). If he buys no Pepsi, he can afford 100 pizzas (point A).



# Figure 1 The Consumer's Budget Constraint



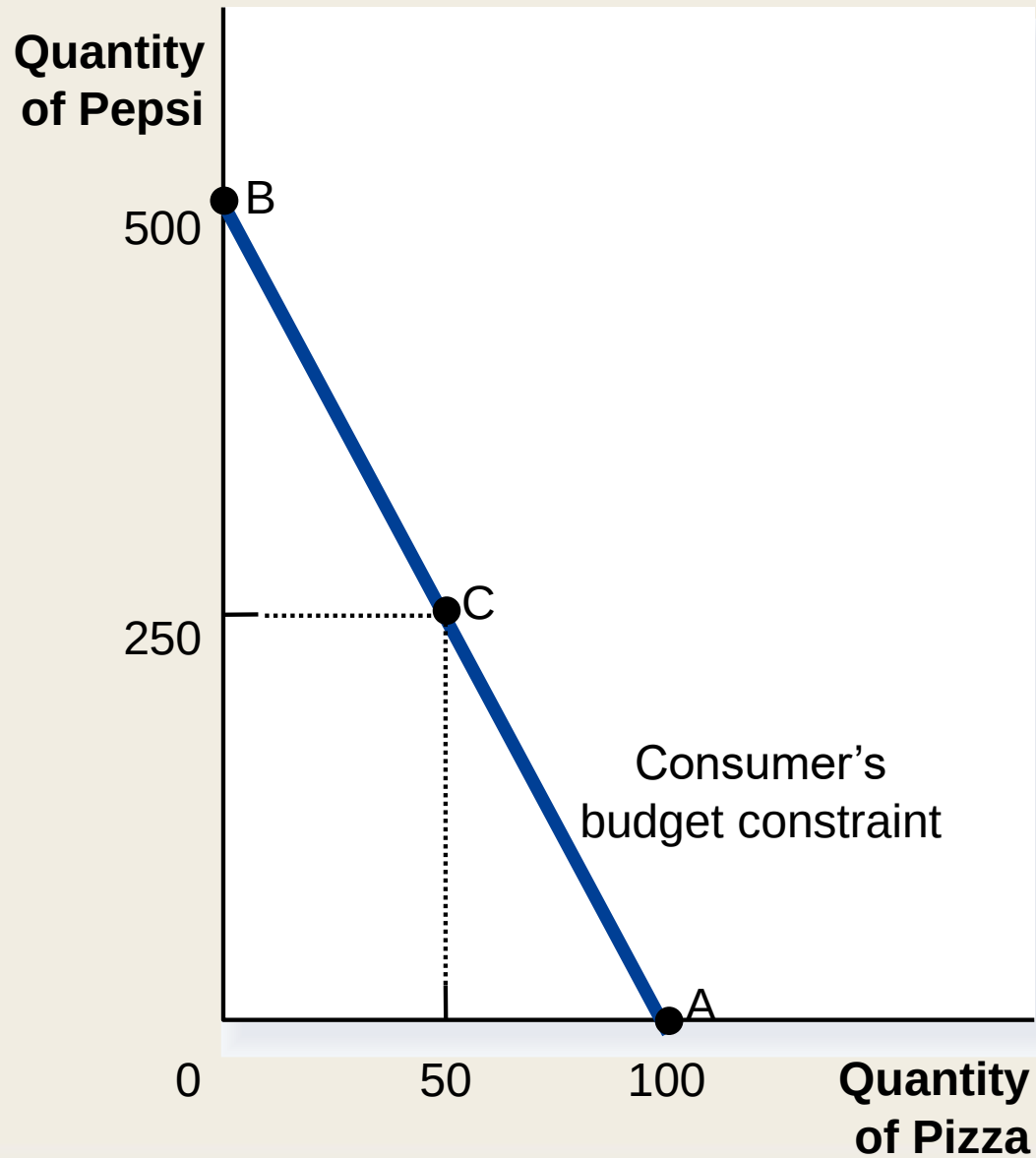




# THE BUDGET CONSTRAINT: WHAT THE CONSUMER CAN AFFORD

- The Consumer's Budget Constraint
  - Alternately, the consumer can buy 50 pizzas and 250 pints of Pepsi.

# Figure 1 The Consumer's Budget Constraint





# THE BUDGET CONSTRAINT: WHAT THE CONSUMER CAN AFFORD

- The *slope* of the budget constraint line equals the relative price of the two goods, that is, *the price of one good compared to the price of the other*.
- It measures the rate at which the consumer can trade one good for the other.



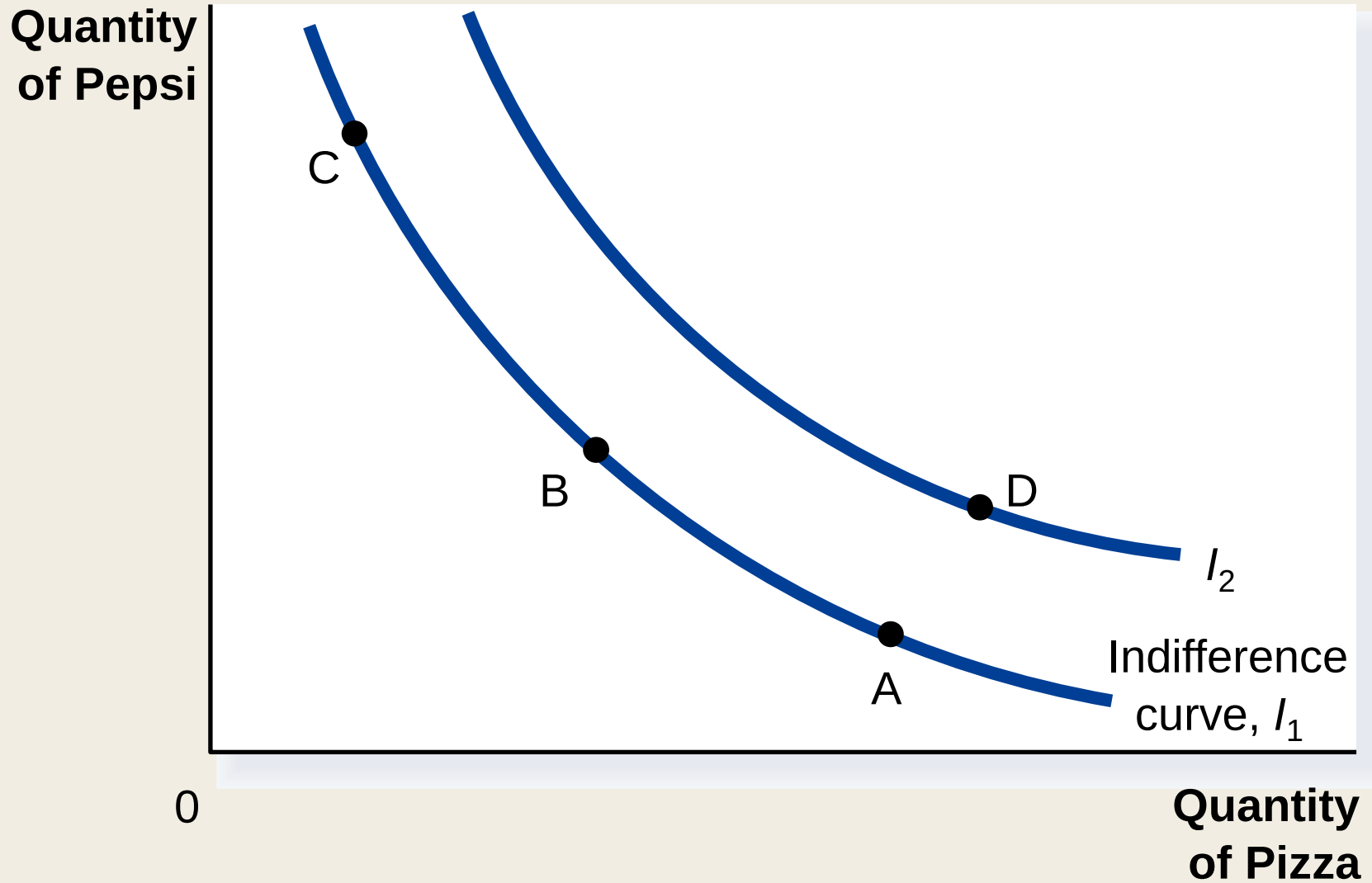
# PREFERENCES: WHAT THE CONSUMER WANTS

- A consumer's preference among consumption bundles may be illustrated with indifference curves.

# Representing Preferences with Indifference Curves

- An *indifference curve* is a curve that shows consumption bundles that give the consumer the same level of satisfaction.

## Figure 2 The Consumer's Preferences

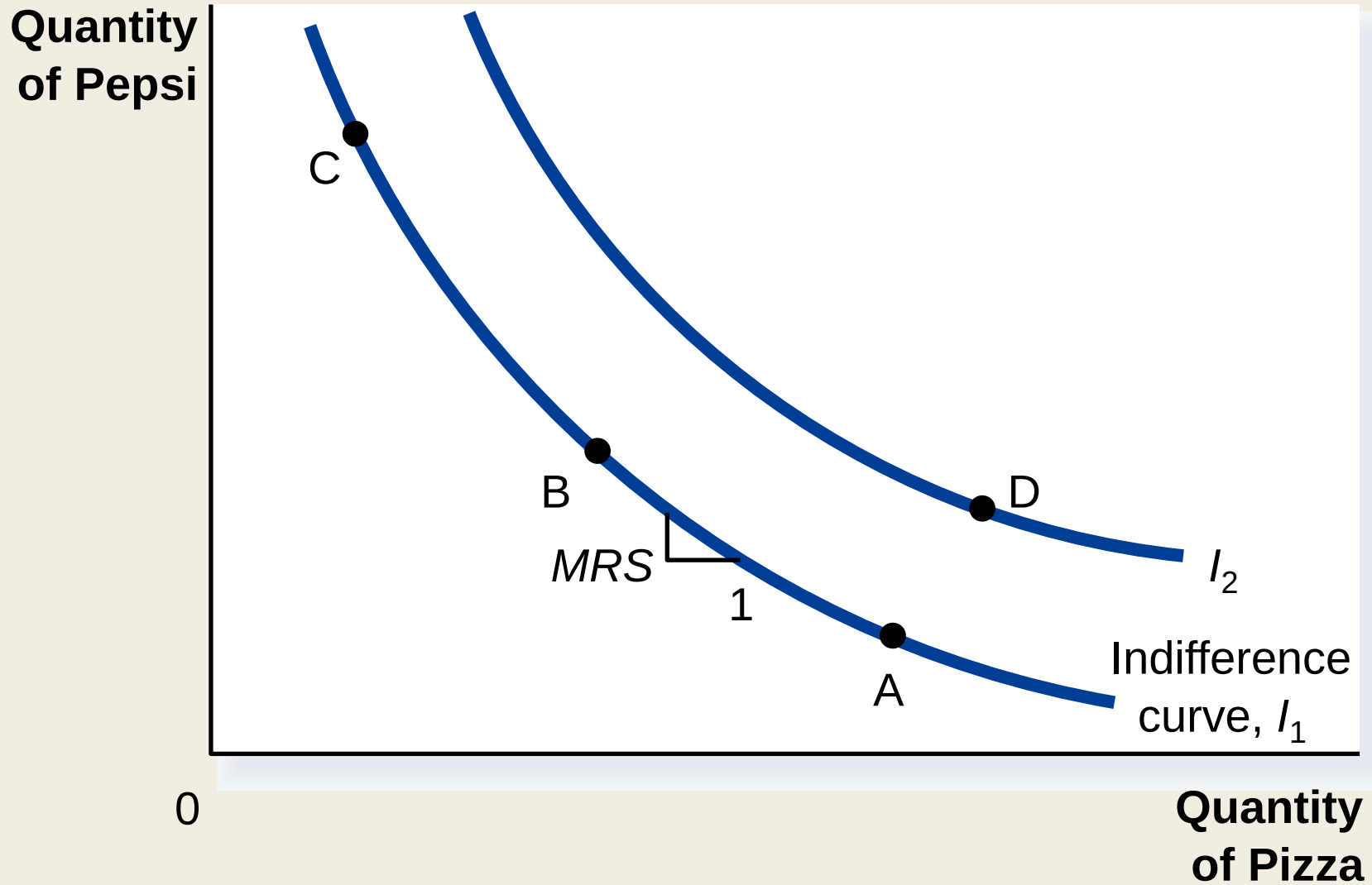


# Representing Preferences with Indifference Curves

- The Consumer's Preferences
  - The consumer is indifferent, or equally happy, with the combinations shown at points A, B, and C because they are all on the same curve.
- The Marginal Rate of Substitution
  - The slope at any point on an indifference curve is the *marginal rate of substitution*.
    - It is the rate at which a consumer is willing to trade one good for another.
    - It is the amount of one good that a consumer requires as compensation to give up one unit of the other good.



## Figure 2 The Consumer's Preferences



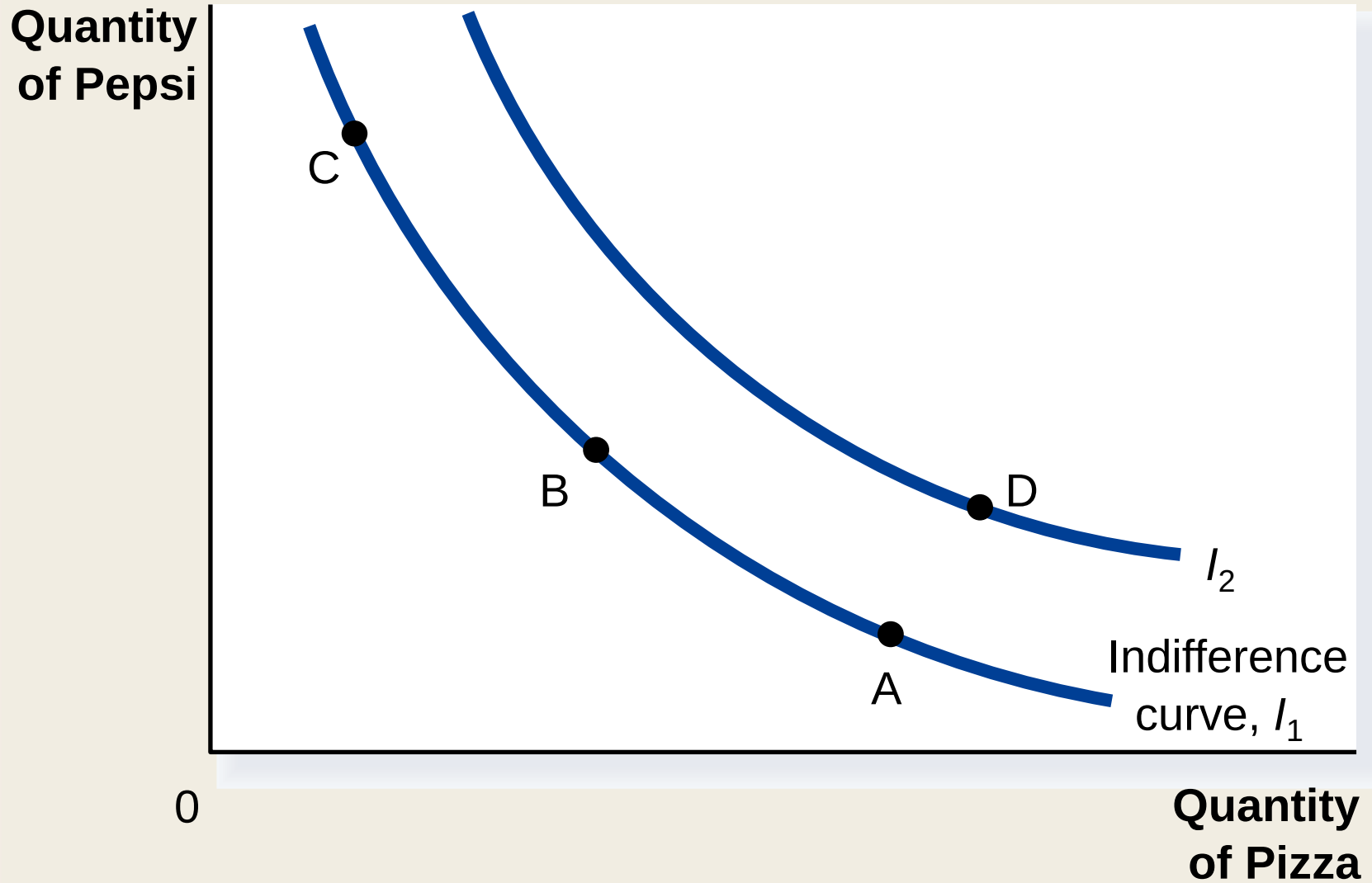
# Four Properties of Indifference Curves

- Higher indifference curves are preferred to lower ones.
- Indifference curves are downward sloping.
- Indifference curves do not cross.
- Indifference curves are bowed inward.

# Four Properties of Indifference Curves

- Property 1: Higher indifference curves are preferred to lower ones.
  - Consumers usually prefer more of something to less of it.
  - Higher indifference curves represent larger quantities of goods than do lower indifference curves.

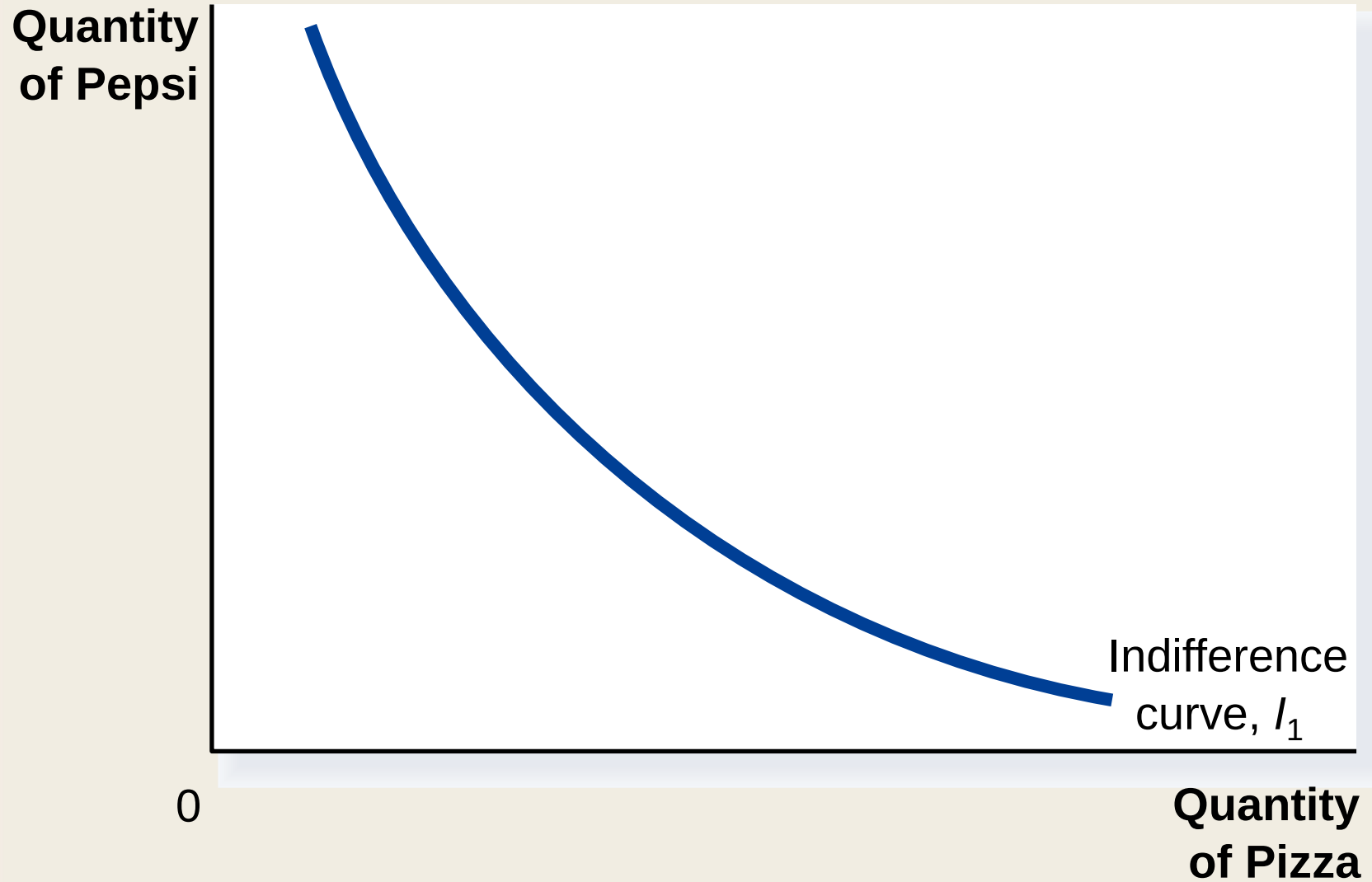
**Figure 2 The Consumer's Preferences**



# Four Properties of Indifference Curves

- Property 2: Indifference curves are downward sloping.
  - A consumer is willing to give up one good only if he or she gets more of the other good in order to remain equally happy.
  - If the quantity of one good is reduced, the quantity of the other good must increase.
  - For this reason, most indifference curves slope downward.
  - Remember, a consumer is equally happy at all points along a given indifference curve.

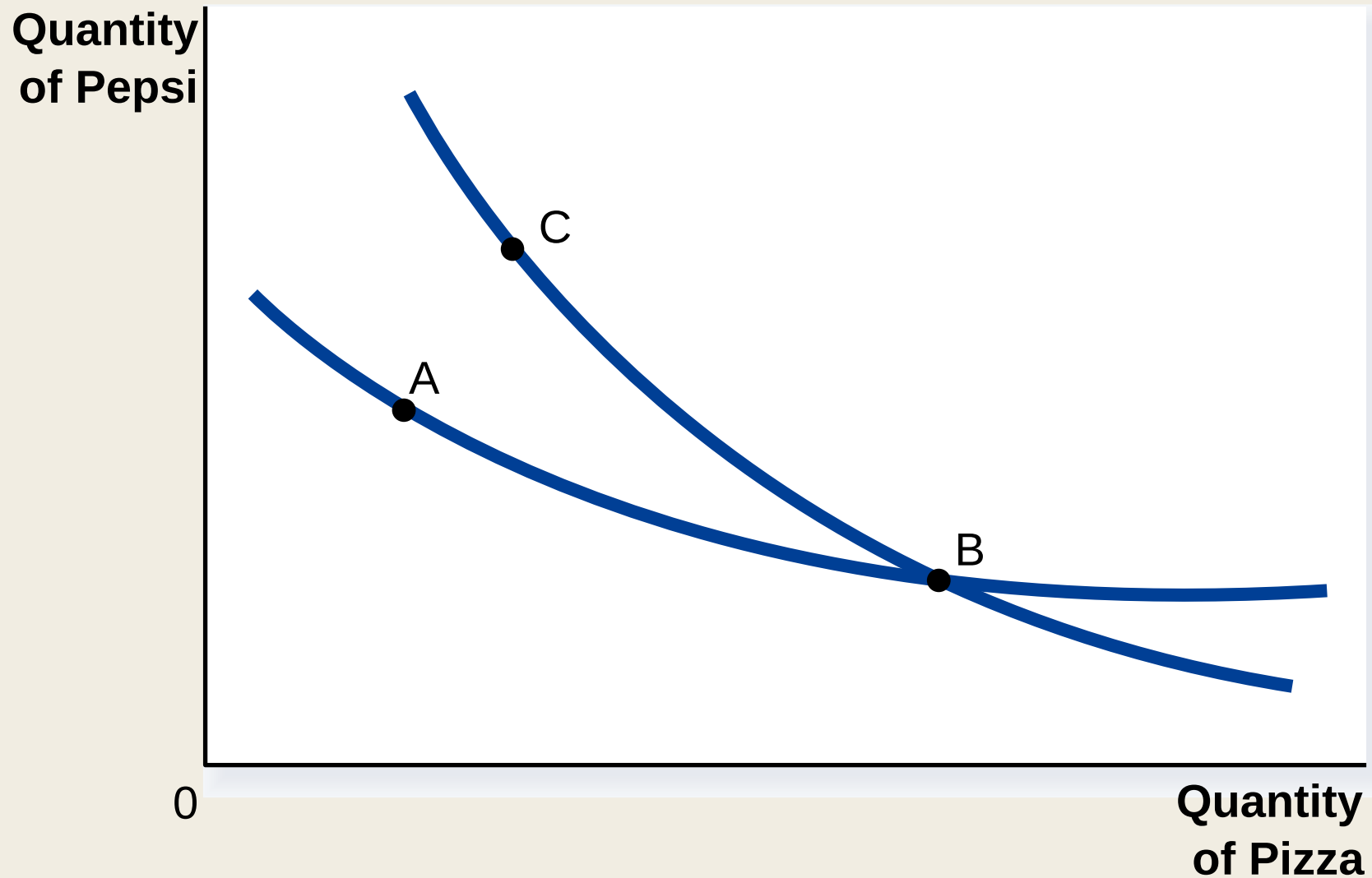
## Figure 2 The Consumer's Preferences



# Four Properties of Indifference Curves

- Property 3: Indifference curves do not cross.
  - Points A and B should make the consumer equally happy.
  - Points B and C should make the consumer equally happy.
  - This implies that A and C would make the consumer equally happy.
  - But C has more of both goods compared to A.

# Figure 3 The Impossibility of Intersecting Indifference Curves

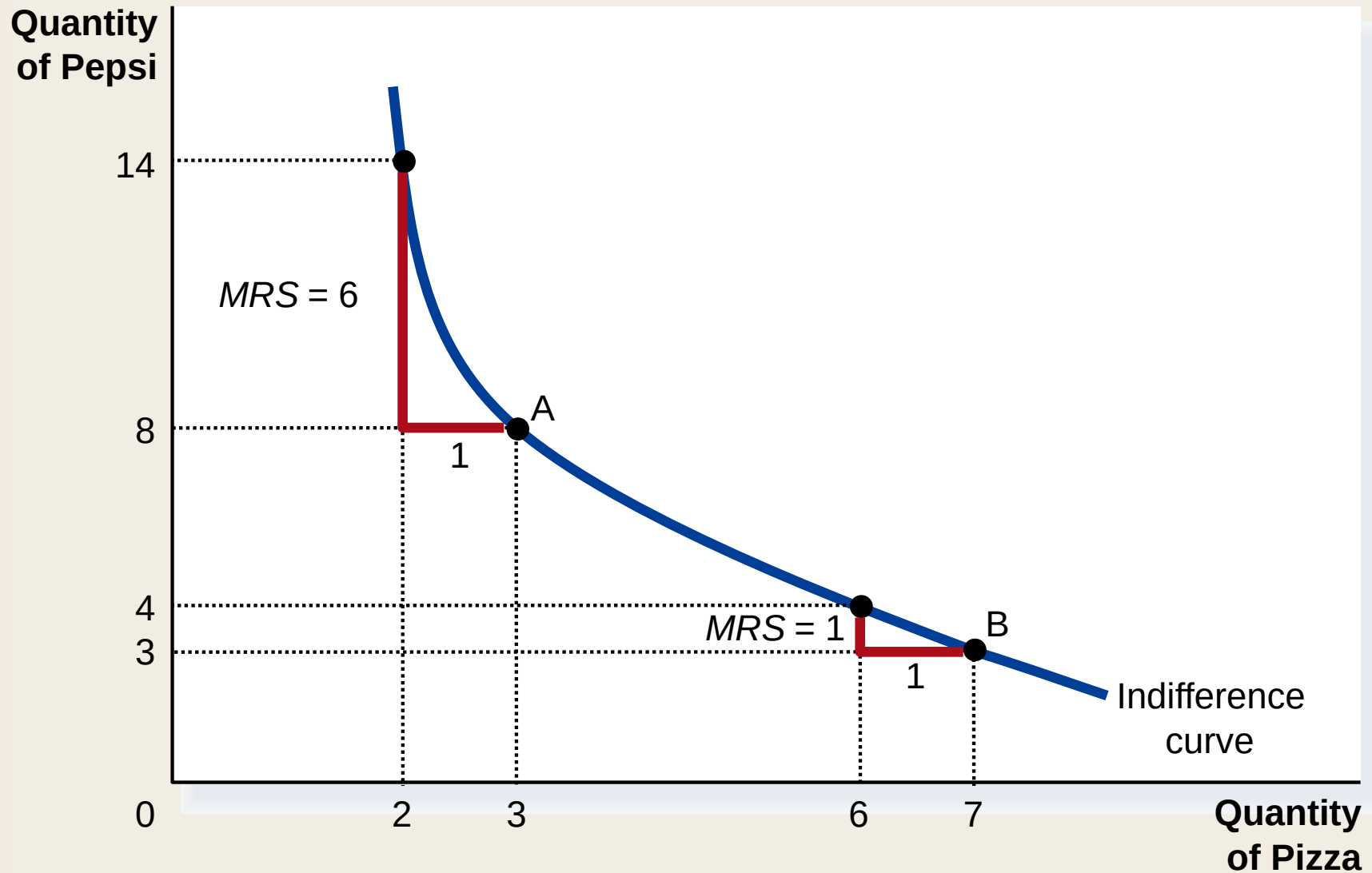




# Four Properties of Indifference Curves

- Property 4: Indifference curves are bowed inward.
  - People are more willing to trade away goods that they have in abundance and less willing to trade away goods of which they have little.
  - These differences in a consumer's marginal substitution rates cause his or her indifference curve to bow inward.

# Figure 4 Bowed Indifference Curves



# Two Extreme Examples of Indifference Curves

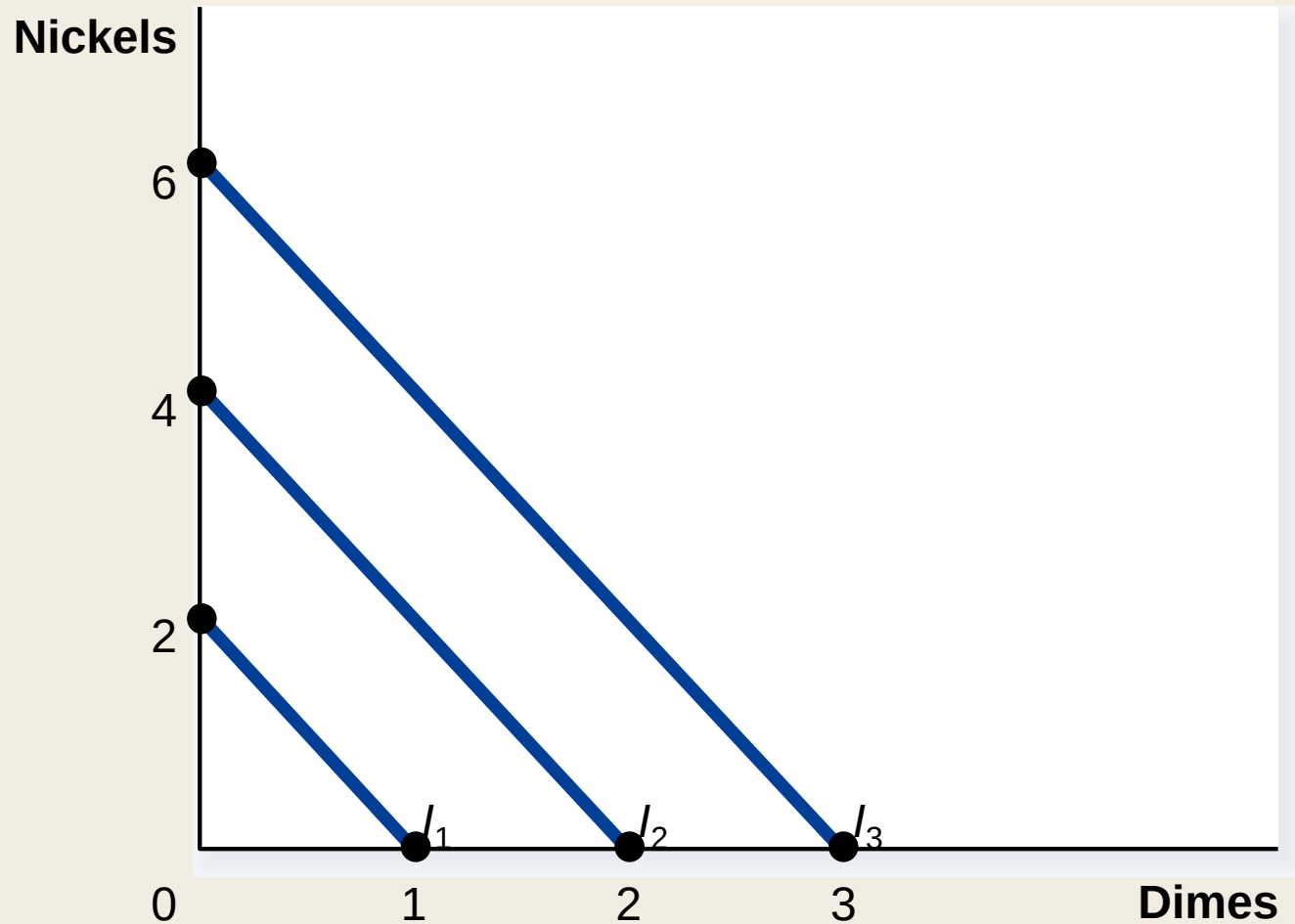
- Perfect substitutes
- Perfect complements

# Two Extreme Examples of Indifference Curves

- Perfect Substitutes
  - Two goods with straight-line indifference curves are *perfect substitutes*.
  - The marginal rate of substitution is a fixed number.

## Figure 5 Perfect Substitutes and Perfect Complements

(a) Perfect Substitutes

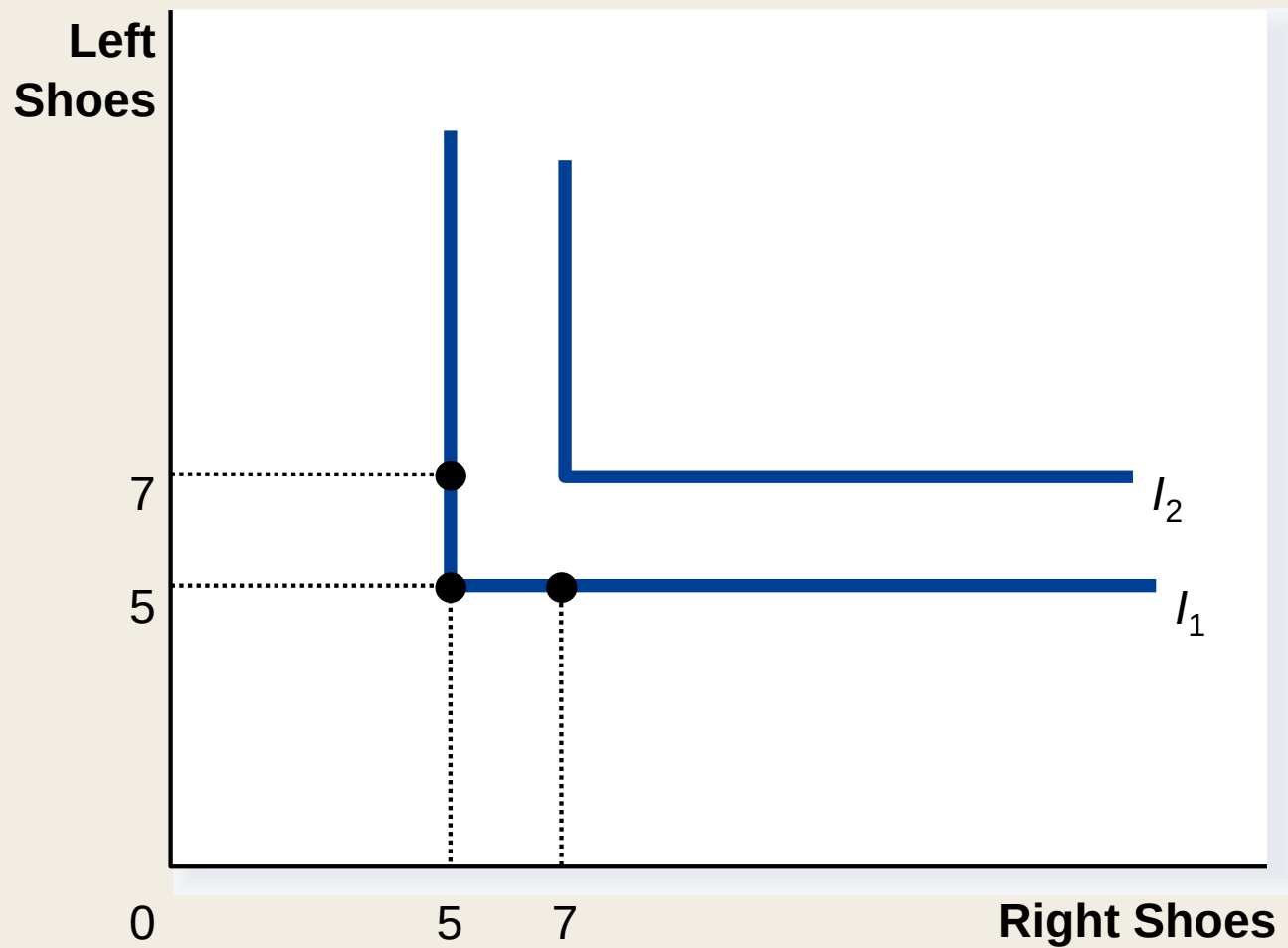


# Two Extreme Examples of Indifference Curves

- Perfect Complements
  - Two goods with right-angle indifference curves are *perfect complements*.
  - Since these goods are always used together, extra units of one good, outside the desired consumption ratio, add no additional satisfaction.

## Figure 5 Perfect Substitutes and Perfect Complements

(b) Perfect Complements





# OPTIMIZATION: WHAT THE CONSUMER CHOOSES

- Consumers want to get the combination of goods on the highest possible indifference curve.
- However, the consumer must also end up on or below his budget constraint.



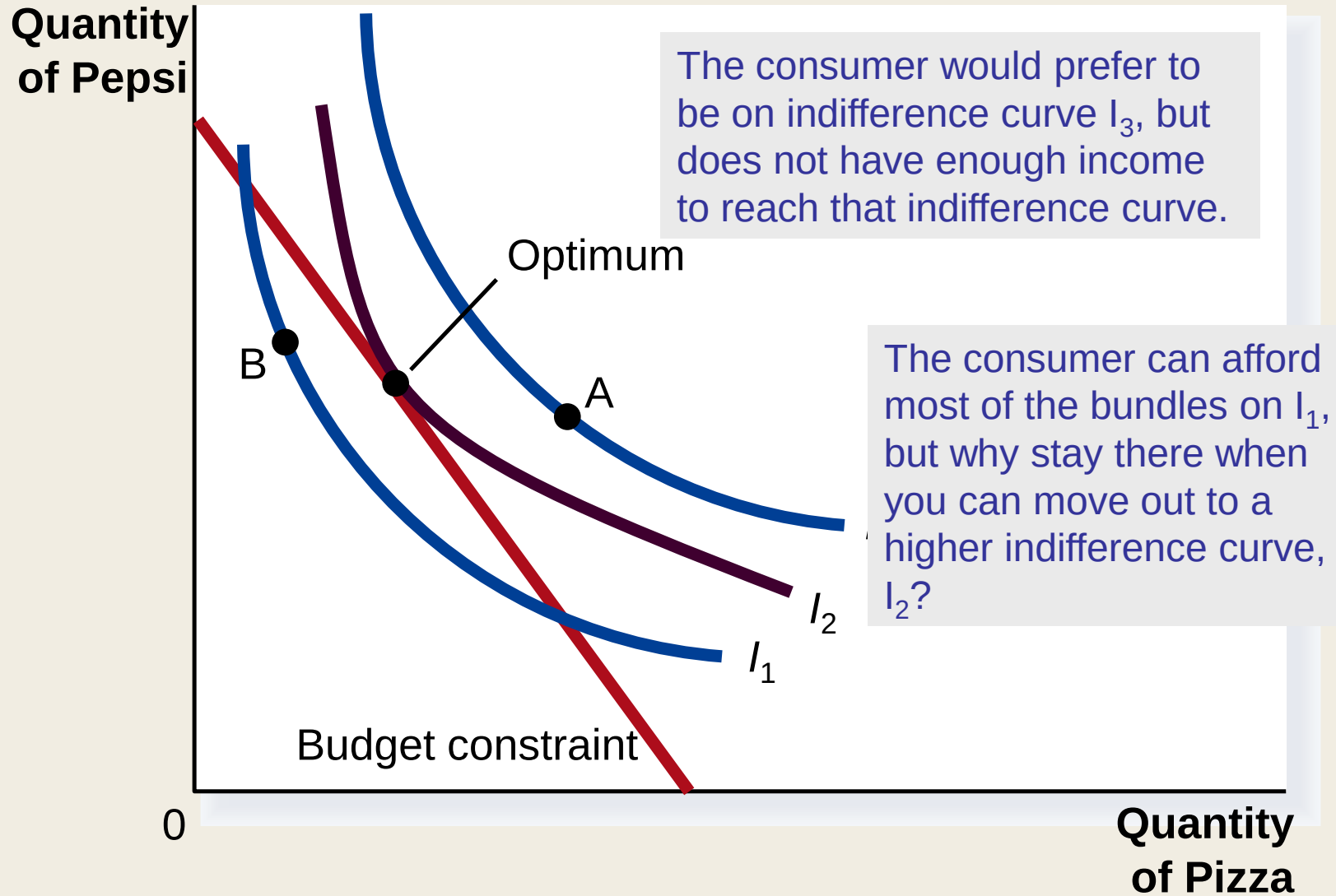
# The Consumer's Optimal Choices

- Combining the indifference curve and the budget constraint determines the consumer's optimal choice.
- Consider a well-behaved indifference curve that is bowed inward.
- Consumer optimum occurs at the point where the highest indifference curve and the budget constraint are tangent.

# The Consumer's Optimal Choice

- The consumer chooses consumption of the two goods so that the marginal rate of substitution equals the relative price.
- At the consumer's optimum, the consumer's valuation of the two goods equals the market's valuation.

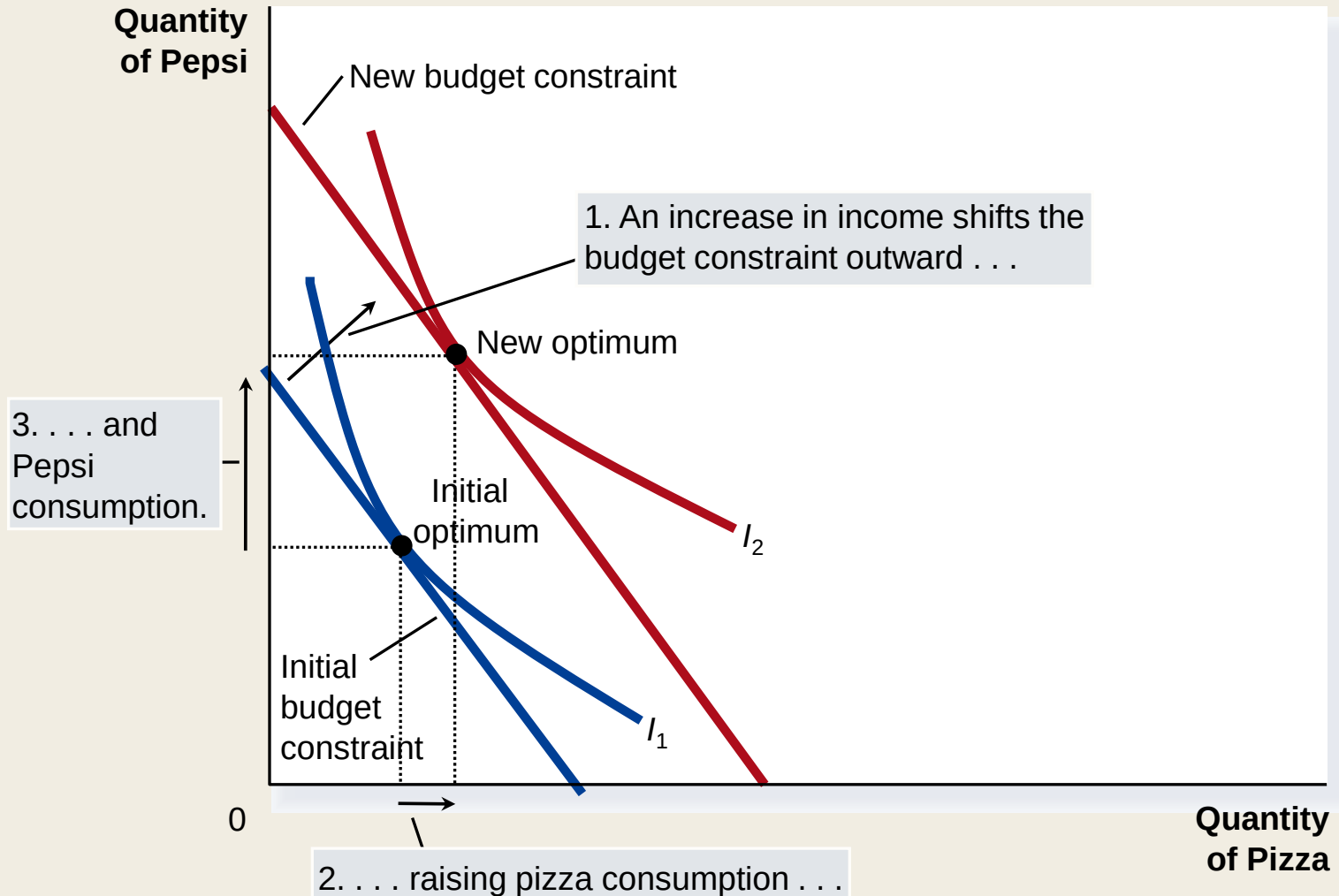
## Figure 6 The Consumer's Optimum



# How Changes in Income Affect the Consumer's Choices

- An increase in income shifts the budget constraint outward.
  - The consumer is able to choose a better combination of goods on a higher indifference curve.

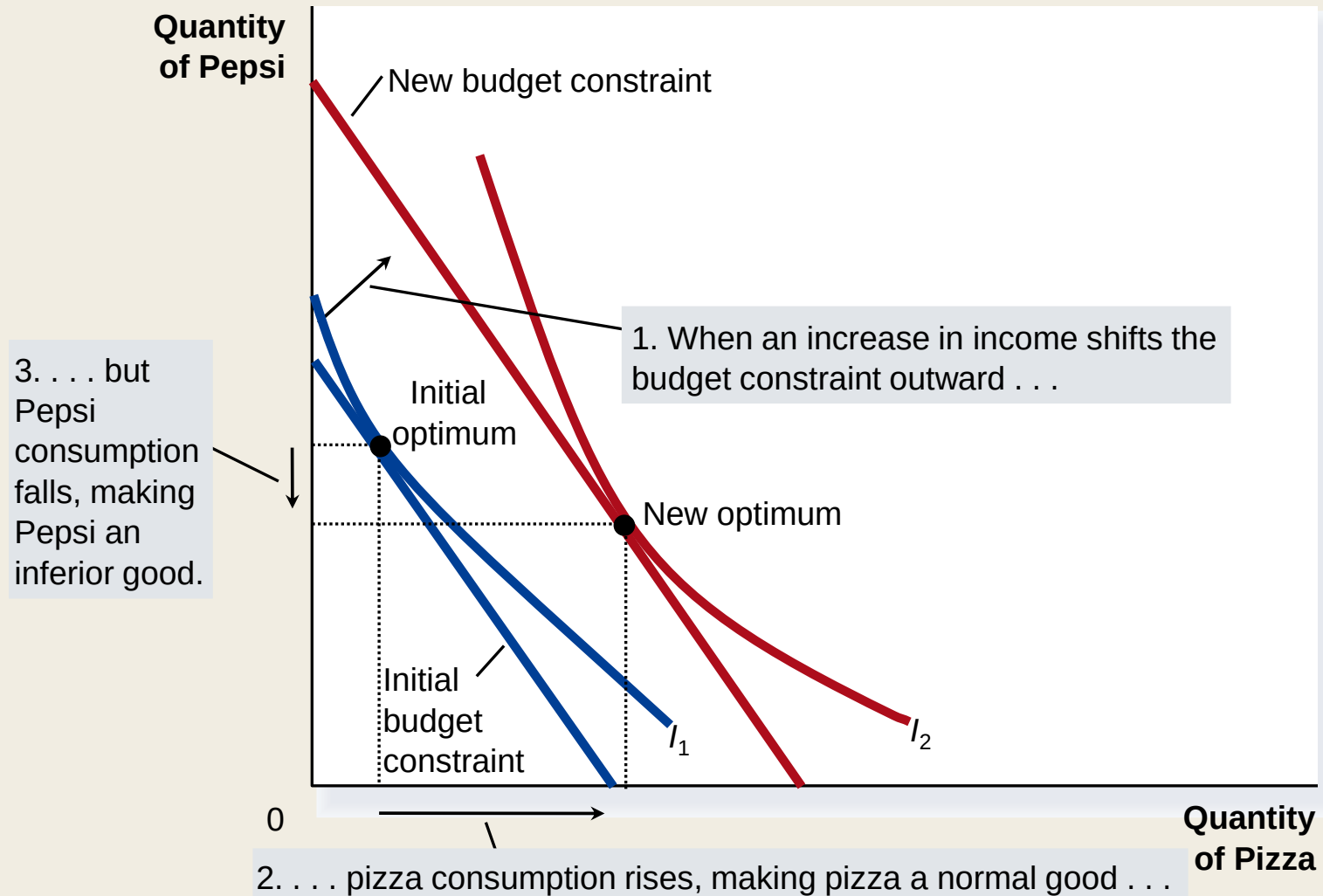
## Figure 7 An Increase in Income



# How Changes in Income Affect the Consumer's Choices

- Normal versus Inferior Goods
  - If a consumer buys more of a good when his or her income rises, the good is called a *normal good*.
  - If a consumer buys less of a good when his or her income rises, the good is called an *inferior good*.

## Figure 8 An Inferior Good

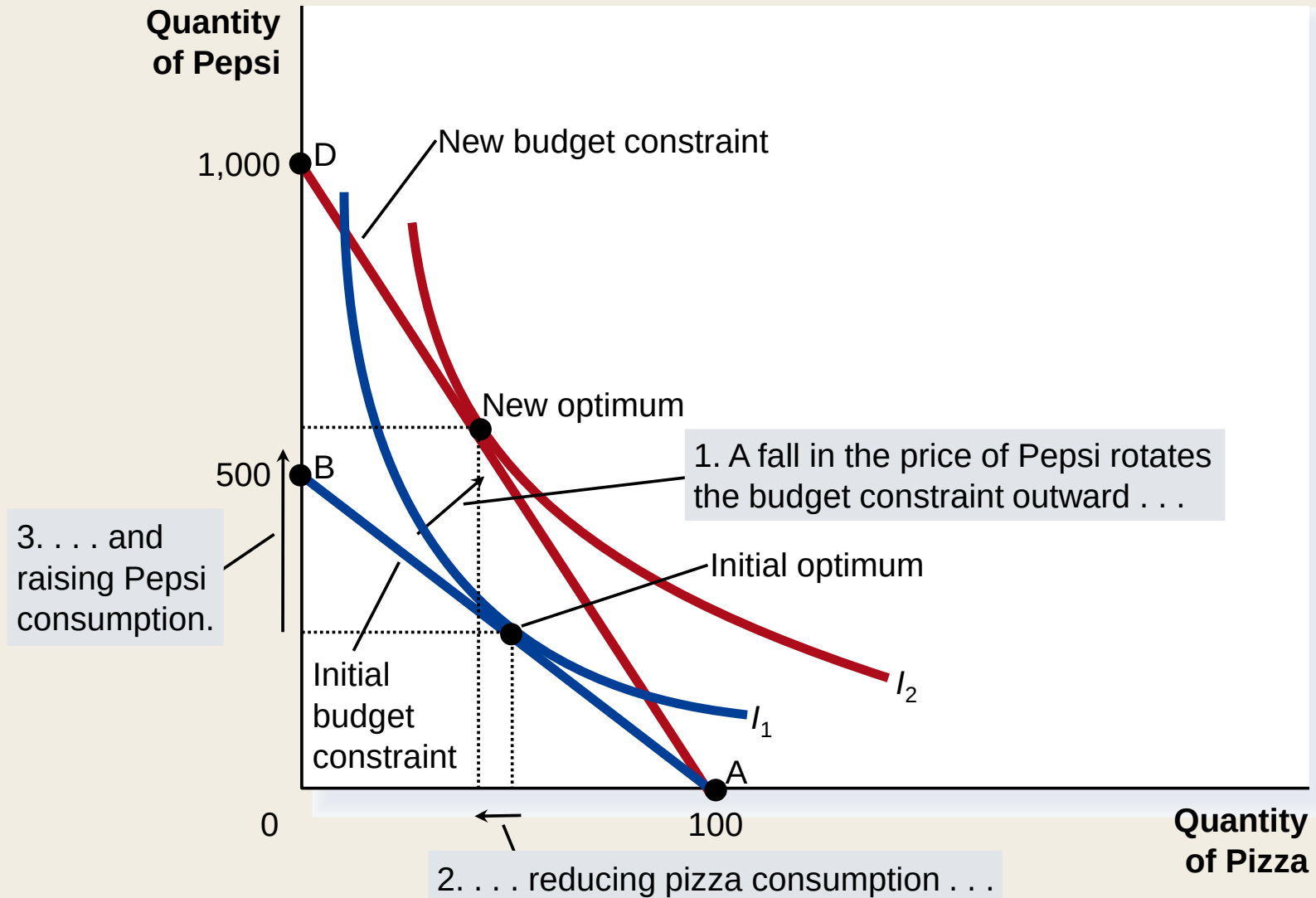


# How Changes in Prices Affect Consumer's Choices

- A fall in the price of any good rotates the budget constraint outward and changes the slope of the budget constraint.



## Figure 9 A Change in Price



# Income and Substitution Effects

- A price change has two effects on consumption.
  - An income effect
  - A substitution effect

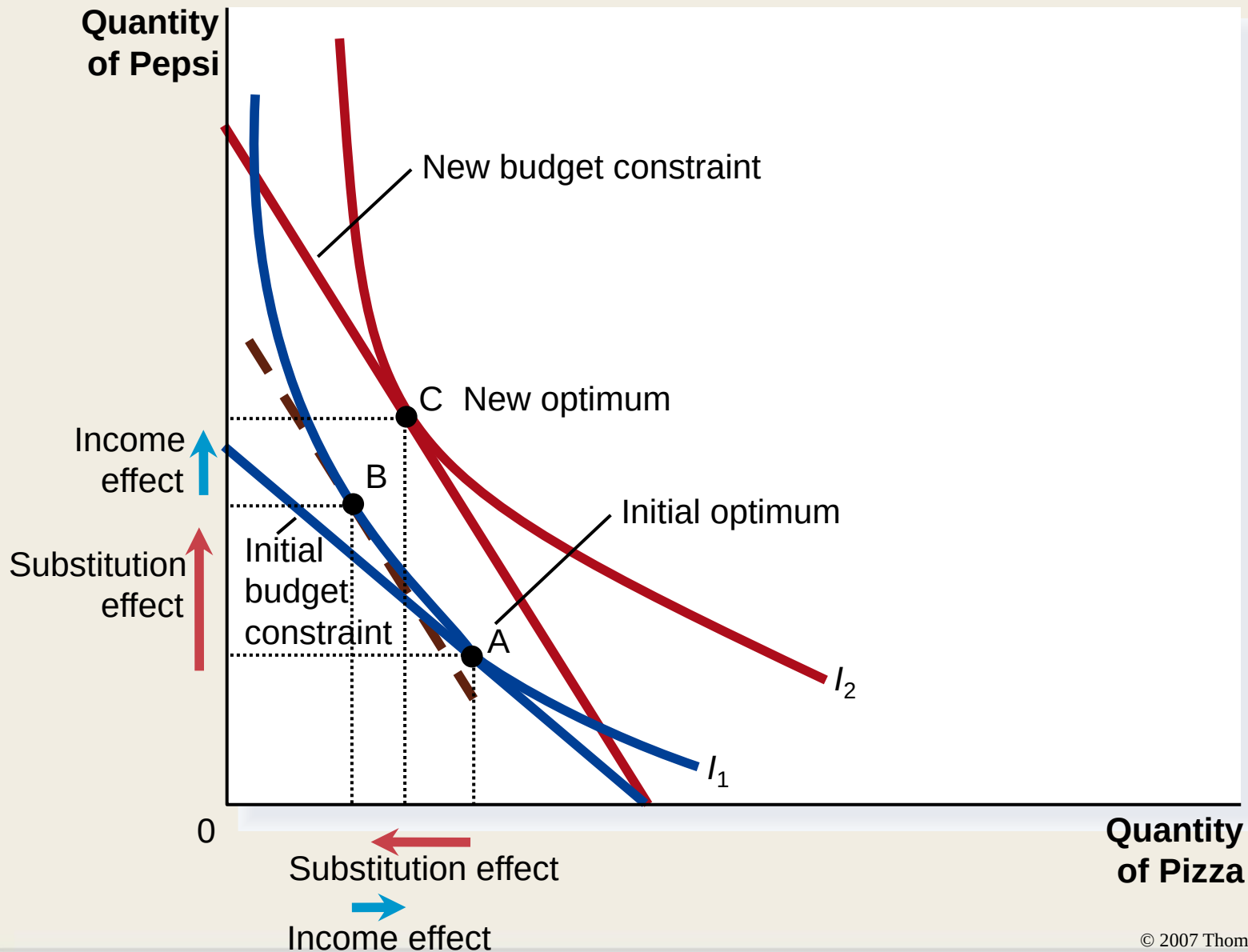
# Income and Substitution Effects

- The Substitution Effect
  - The *substitution effect* is the change in consumption that results when a price change moves the consumer along an indifference curve to a point with a different marginal rate of substitution.
- The Income Effect
  - The *income effect* is the change in consumption that results when a price change moves the consumer to a higher or lower indifference curve.

# Income and Substitution Effects

- A Change in Price: Substitution Effect
  - A price change first causes the consumer to move from one point on an indifference curve to another on the same curve.
    - Illustrated by movement from point A to point B.
- A Change in Price: Income Effect
  - After moving from one point to another on the same curve, the consumer will move to another indifference curve.
    - Illustrated by movement from point B to point C.

# Figure 10 Income and Substitution Effects



## Table 1 Income and Substitution Effects When the Price of Pepsi Falls

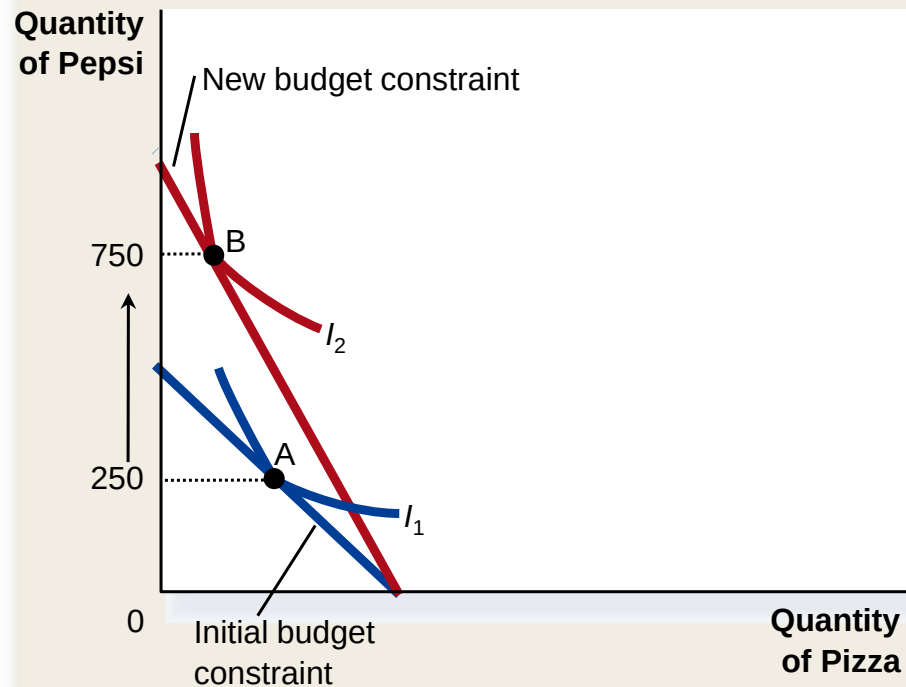
| Good  | Income Effect                              | Substitution Effect                                              | Total Effect                                                                                                       |
|-------|--------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Pepsi | Consumer is richer, so he buys more Pepsi. | Pepsi is relatively cheaper, so consumer buys more Pepsi.        | Income and substitution effects act in same direction, so consumer buys more Pepsi.                                |
| Pizza | Consumer is richer, so he buys more pizza. | Pizza is relatively more expensive, so consumer buys less pizza. | Income and substitution effects act in opposite directions, so the total effect on pizza consumption is ambiguous. |

# Deriving the Demand Curve

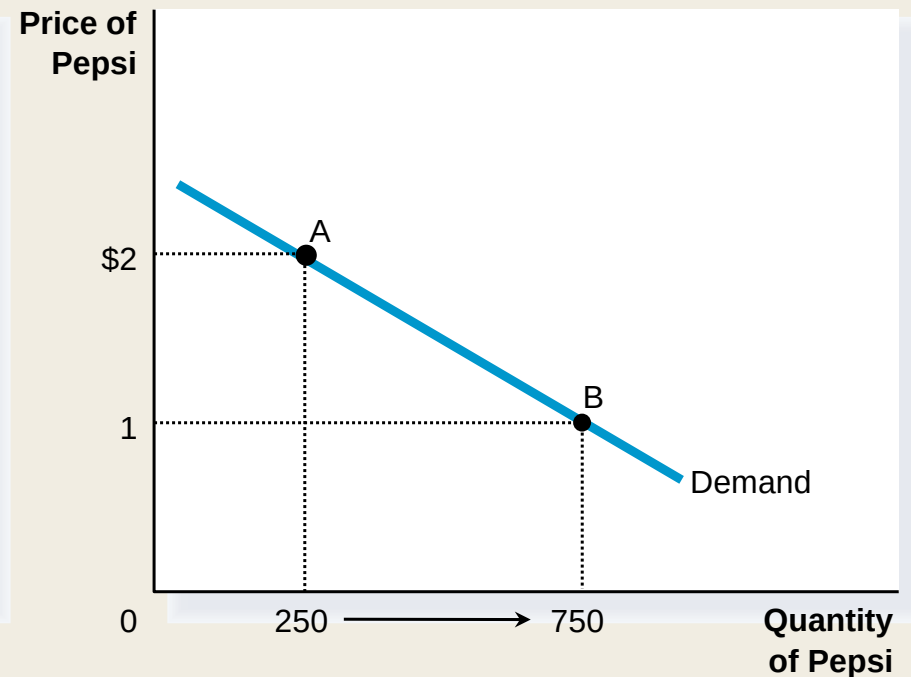
- A consumer's demand curve can be viewed as a summary of the optimal decisions that arise from his or her budget constraint and indifference curves.

# Figure 11 Deriving the Demand Curve

(a) The Consumer's Optimum



(b) The Demand Curve for Pepsi







# THREE APPLICATIONS

- Do all demand curves slope downward?
- How do wages affect labor supply?

# Do All Demand Curves Slope Downward?

- Demand curves can sometimes slope upward.
- This happens when a consumer buys more of a good when its price rises.
- Suppose there is a commodity  $x$  and its price is denoted by  $P$ .
- When  $P$  falls,  $x$  becomes relatively cheaper in relation to the other good  $y$ .

- Substitution effect entails the consumer will consume more of  $x$  even though he is confined to the same indifference curve of the pre price change level. This holds for both normal and inferior goods.
- For normal goods, income effect will entail an increase in consumption of  $x$ .
- For inferior goods, income effect will entail a decrease in consumption of  $x$ .

- For normal goods, both SE and IE work in the same direction and consumption of  $x$  is increased.
- As a result, the demand curve for normal goods is always downward sloping.

- For inferior goods, SE and IE work in opposite directions. According to SE,  $x$  is raised. But according to IE,  $x$  is decreased.
- If SE dominates IE, then for such inferior goods, the demand curve is downward sloping.
- If IE dominates SE, then for such inferior goods, the demand curve is upward sloping.

- *Giffen goods*

- Economists use the term Giffen good to describe an inferior good that violates the law of demand, where the demand curve is upward sloping.
- Giffen goods are goods for which an increase in the price raises the quantity demanded.

# How Do Wages Affect Labor Supply?

- Suppose there are two goods: Consumption and Leisure.
- The price of the consumption good is  $P$ , while the price (opportunity cost) of Leisure is Wage ( $w$ ).
- Suppose  $w$  goes up.
- Leisure becomes relatively more expensive.
- SE dictates reduction of leisure.

- Note that even after consuming the initial (Consumption, Leisure) bundle at the post-change scenario with higher  $w$ , the consumer will now have a higher income.
- This will generate an income effect, according which a higher income will increase leisure.



- SE and IE work in opposite directions:
- According to SE, when wage rises leisure should be reduced
- According to IE, when wage rises leisure should be increased

- If the substitution effect is greater than the income effect for the worker, he or she reduces leisure and therefore works more.
- If income effect is greater than the substitution effect, he or she increases leisure and therefore works less.

# Summary

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- A consumer's budget constraint shows the possible combinations of different goods he can buy given his income and the prices of the goods.
- The slope of the budget constraint equals the relative price of the goods.
- The consumer's indifference curves represent his preferences.

# Summary

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- Points on higher indifference curves are preferred to points on lower indifference curves.
- The slope of an indifference curve at any point is the consumer's marginal rate of substitution.
- The consumer optimizes by choosing the point on his budget constraint that lies on the highest indifference curve.

# Summary

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- When the price of a good falls, the impact on the consumer's choices can be broken down into an income effect and a substitution effect.
- The income effect is the change in consumption that arises because a lower price makes the consumer better off.
- The income effect is reflected by the movement from a lower to a higher indifference curve.

# Summary

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- The substitution effect is the change in consumption that arises because a price change encourages greater consumption of the good that has become relatively cheaper.
- The substitution effect is reflected by a movement along an indifference curve to a point with a different slope.

# Summary

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- The theory of consumer choice can explain:
  - Why demand curves can potentially slope upward.
  - How wages affect labor supply.